

Paper A/B Target Submission Packet

Exemplar and Guideline Manifest

Date: 2026-06-12

Scope: target-journal and exemplar-paper evidence used to reshape Paper A and Paper B into near-submission manuscript packages. Closed PDFs obtained through PSU institutional access may be used locally, but they are not redistributed in Git or OSF.

Source	Access state	Use in current package
Computers & Education Guide for Authors	Web verified	Paper A target-journal package: abstract, keywords, highlights, editable tables, double-anonymized file separation
Research Synthesis Methods author instructions	Web verified	Paper B target-journal package: 250-word abstract, RSM highlights, repository/data-availability expectations, GenAI evaluation transparency
Scherer, Siddiq, and Tondeur (2019), Computers & Education	Bibliographic record verified; closed PDF needed for exact table/figure replication if not locally available	Primary Paper A exemplar for TAM/MASEM positioning in C&E
Or (2024), OTESSA Journal	PDF extracted locally, 26 pages	Open OSMASEM/TAM education exemplar for methods, PRISMA/table/figure spine
Schmidt et al. (2024), ALTARS/arXiv	PDF extracted locally, 30/32 pages	Paper B exemplar for manual evaluation, domain-specific data extraction accuracy, and caution against autonomous replacement
Schroeder et al. (2025), arXiv AIDE/HIL	PDF extracted locally, 12 pages	Paper B exemplar for explicit versus derived variable families, human-in-the-loop workflow, and source panel/provenance framing
Legate and Nimon F1000 protocol	Local PDF extracted, 28 pages	Paper B social-science methods gap and living-review/reporting workflow context
RSM GenAI submission guidance	Web/PMC/RSM metadata verified	Paper B reporting benchmark for validation, prompt/model versioning, locked outputs, and transparent AI declaration
Annals AI-assisted data-extraction SWAR	Web metadata verified; publisher PDF blocked	Comparator for workflow validation, time/error outcome reporting, and human verification framing

Source Links

- Computers & Education guide: <https://www.sciencedirect.com/journal/computers-and-education/publish/guide-for-authors>
- Research Synthesis Methods instructions: <https://www.cambridge.org/core/journals/research-synthesis-methods/information/author-instructions/preparing-your-materials>
- Scherer et al. DOI: <https://doi.org/10.1016/j.compedu.2018.09.009>
- Or (2024) DOI: <https://doi.org/10.18357/otessaj.2024.4.3.66>
- Schmidt et al. arXiv: <https://arxiv.org/abs/2405.14445>
- Schroeder et al. arXiv: <https://arxiv.org/abs/2501.11840>
- RSM automated meta-analysis review: <https://doi.org/10.1017/rsm.2025.10065>
- RSM GenAI submission guidance: <https://doi.org/10.1017/rsm.2025.10058>
- Annals prospective cohort DOI: <https://doi.org/10.7326/ANNALS-25-00739>
- OSF repository for Paper B: <https://osf.io/mkrgd/overview>

Exemplar Decomposition Matrix

Date: 2026-06-12

Transferable Patterns

Exemplar	Method spine	Table/Figure pattern	How to use it here	Boundary
Scherer et al. C&E TAM- MASEM	Education technology adoption framed around TAM, MASEM, theory competition	PRISMA, study-characteristics table, pooled-matrix/path tables, conceptual/path figures	Use as Paper A target-journal model for C&E positioning and MASEM result architecture	Need PSU PDF for exact visual replication before final submission
Or 2024 OSMASEM	One-step MASEM of TAM in education using R/metaSEM references	PRISMA/selection flow, source-study table, model fit table, path/effect table, model diagrams	Use as Paper A open-PDF reference for OSMASEM reporting and figure spine	Open exemplar is useful but not the target C&E house style
Schmidt et al. 2024	Prompt development plus manual evaluation by study domain and extraction item	Prompt table, extraction-variable table, accuracy-by-domain/item tables, cautionary narrative	Use for Paper B task-family denominators and manual scoring explanation	Do not import one overall accuracy denominator
Schroeder et al. 2025	Pilot-to-full human-in-the-loop extraction across explicit and derived variables	Workflow/application figures, exact/accurate match tables, model result table	Use for Paper B explicit versus derived task-family reporting and HIL figure	Use as workflow comparator, not a source for replacement claims
F1000 social-	Living review of semi-	Workflow and reporting	Use for Paper B social-	Protocol status means

Exemplar	Method spine	Table/Figure pattern	How to use it here	Boundary
science protocol	automated extraction approaches in social sciences	figures, protocol tables, agreement/reliability plan	science contribution and broader evidence-synthesis relevance	use for rationale, not result benchmarking
RSM GenAI guidance	Reporting expectations for GenAI evaluations in SRMA	Validation/ methodology/ transparency checklist logic	Use to structure Paper B methods, AI declaration, model/prompt reproducibility, repository statement	RSM expects clear model-version and use-case boundary
Annals SWAR	Prospective workflow validation with human verification	Concordance, accuracy, time-on-task, error severity tables	Use for Paper B discussion of workflow evaluation outcomes	Publisher PDF needed for exact table/figure deconstruction

Design Consequences

Paper A should look like a C&E empirical synthesis article: concise education-technology contribution, theory-forward introduction, transparent PRISMA and coding workflow, then pooled matrices, path models, moderators, and sensitivity diagnostics.

Paper B should look like an RSM methods article: it should foreground validation design, source-anchored reference construction, denominator-family scoring, locked model outputs, reproducible scripts, and claim boundaries. The central claim is workflow augmentation and review triage, not LLM replacement.

Paper A Target-Journal Structure Map

Target journal: Computers & Education

Decision state: target journal, construct scope, figure spine, and table spine are approved. The actual TSSEM/OSMASEM/sensitivity estimate insertion is approved but cannot be treated as complete from the current input because the model-ready primary file lacks source-supported numeric sample size for most rows.

Current Analysis Gate

Check	Current value	Implication
Primary model-ready rows	804	Available in the OneDrive Paper1 working folder
Usable 10-construct rows after r checks	796	No r absolute value ≥ 1 blocker in the tiered primary file
Rows missing numeric N	754	N-weighted TSSEM/MASEM cannot be claimed until sample size is reconciled or explicit exclusion rule is applied

Check	Current value	Implication
Studies represented	74	Correlation evidence is spread across incomplete matrices
Covered construct pairs	44/45	Coverage is broad but not complete
Complete 10-construct matrices	0	Complete-case 10-construct TSSEM is not feasible as the primary route
Studies with 15 or more construct pairs	26	FIML/TSSEM route may be possible after N reconciliation

Least-covered pairs:

Construct pair	Rows
SE-TRU	1
ANX-FC	2
ANX-SE	2
ANX-UB	2
ATT-TRU	2
ANX-BI	3
ANX-EE	3
ANX-SI	3
ANX-PE	4
FC-TRU	4
ATT-SE	5
SE-UB	5

C&E Submission Components

- Main manuscript: double-anonymized Word file with title page separated at submission.
- Abstract: no more than 250 words.
- Keywords: 1 to 7.
- Highlights: 3 to 5 bullets, no more than 85 characters each, submitted separately.
- Tables: editable text, cited in order, with captions and notes.
- Figures: conceptual model, PRISMA, coverage heatmap, and path model as editable/high-resolution files.

Proposed Highlights

- Synthesizes AI adoption evidence in higher education with MASEM.
- Integrates TAM/UTAUT predictors with trust and AI anxiety.
- Separates direct-r inputs from converted sensitivity evidence.
- Tests structural and moderator paths across ten constructs.
- Provides reproducible extraction and QC artifacts.

Section Spine

1. Introduction: higher-education AI adoption problem, why structural synthesis is needed, why AI trust/anxiety expand TAM/UTAUT.
2. Theory and hypotheses: TAM/UTAUT, attitude mediation, self-efficacy, trust, anxiety, use behavior.
3. Method: search, screening, source adjudication, coding, 10-construct harmonization, TSSEM/OSMASEM, sensitivity.
4. Results: PRISMA, study characteristics, input coverage, Stage 1 pooled matrix, Stage 2 model paths, moderators, sensitivity.
5. Discussion: theory, institutional implications, limitations, reproducibility, future AI-adoption measurement.

Table Spine

Table	Title	Status
Table 1	Construct harmonization and operational definitions	Ready to draft
Table 2	PRISMA and study-characteristics profile	Needs final inclusion lock
Table 3	Analysis-ready input sets and source-type rules	Ready with current caveats
Table 4	Construct-pair coverage and missingness	Ready as pre-model table
Table 5	Stage 1 pooled correlation matrix	Needs TSSEM run after N gate
Table 6	Stage 2 structural paths and indirect effects	Needs TSSEM run
Table 7	Moderator and sensitivity results	Needs OSMASEM/sensitivity run

Figure Spine

Figure	Purpose	Status
Figure 1	PRISMA flow from 22,166 records to final included studies	Needs final inclusion lock
Figure 2	Ten-construct conceptual model	Ready to draw
Figure 3	Construct-pair coverage heatmap	Ready from current coverage

Figure	Purpose	Status
Figure 4	Final Stage 2 path model	Needs TSSEM run
Figure 5	Moderator/sensitivity comparison	Needs OSMASEM/sensitivity run

Manuscript Boundary

Do not insert final C&E results claims until numeric N is reconciled or an approved N-eligible subset rule is documented for Paper A. The currently defensible manuscript state is introduction, theory spine, methods, input/QC results, and analysis gate.

Paper B Target-Journal Structure Map

Target journal: Research Synthesis Methods

Decision state: Paper B is a source-anchored, human-adjudicated LLM augmentation/validation study with downstream MASEM/TSSEM diagnostic checks. It is not an LLM replacement paper and not a vendor-ranking benchmark.

Current Full-Corpus Gate

Check	Current value	Implication
Full-corpus Step 5 shell exists	True	The 2,043-row target shell can anchor the next M1-R run
Task rows in shell	2043	Full-corpus M1-R target denominator before exception handling
Private source packets present	10/194 expected studies	Current inherited workspace lacks the full private packet set
Bounded M1-R shard	90 rows completed	Diagnostic only
Post-freeze reference	213 studies frozen on 2026-06-09	Governing reference layer
OSF archive	https://osf.io/mkrgd/overview	Share-safe public repository exists

Denominator-family shell counts:

Family	Rows
primary_latent_or_construct_correlation_with_source_type_flag	931
primary_direct_r_or_source_reported_correlation	697
secondary_beta_or_path_converted_effect_size	415

RSM Submission Components

- Abstract: no more than 250 words and readable to a multidisciplinary audience.
- Keywords: 4 to 6, plus RSM-specific keywords at submission.
- Required highlights in manuscript: What is already known; What is new; Potential impact for Research Synthesis Methods readers.
- Data availability statement with repository DOI/URL or explanation.
- AI-methods transparency: model names/versions, dates, access path, prompts, validation, and output-locking process.

Proposed RSM Highlights

What is already known

Systematic review data extraction is labor-intensive, and LLMs can sometimes support extraction workflows. Existing studies show promising but task-dependent accuracy and require human verification.

What is new

This study validates a locked LLM workflow against a source-anchored adjudicated human reference standard for MASEM-ready extraction. It reports task-family denominators, source conditions, human-disagreement traces, and downstream substitution diagnostics rather than one pooled accuracy score.

Potential impact for Research Synthesis Methods readers

The design shows how to evaluate LLM extraction as auditable workflow augmentation for complex evidence synthesis, especially when downstream meta-analytic models depend on numeric source accuracy and sample-size eligibility.

Table Spine

Table	Title	Status
Table 1	Data states and claim roles	Ready
Table 2	Reference construction and source-adjudication workflow	Ready
Table 3	Task-family denominators and scoring rules	Ready
Table 4	Locked model outputs and coverage	Ready for legacy; full-corpus gate pending
Table 5	RQ1 extraction validity by denominator family	Ready with legacy package and bounded diagnostic caveat
Table 6	RQ2 error taxonomy by source condition	Ready with legacy package
Table 7	RQ3 review-priority triage	Ready with legacy package
Table 8	Post-freeze full-corpus M1-R expansion gate	Ready as gate, not result
Table 9	Downstream substitution and TSSEM diagnostics	Ready as bounded subset diagnostic

Figure Spine

Figure	Purpose	Status
Figure 1	Five-step source-anchored validation workflow	Ready
Figure 2	Task-family scoring architecture	Ready
Figure 3	Locked-output and model-provenance flow	Ready
Figure 4	Accuracy/abstention profile by denominator family	Ready with caveats
Figure 5	Source-risk and human-disagreement triage heatmap	Ready
Figure 6	Downstream substitution diagnostic	Ready as bounded subset only

Manuscript Boundary

Permitted: Paper B can claim a reproducible, source-anchored workflow for evaluating LLM-assisted extraction by task family, and it can report bounded diagnostic evidence that the expert-reviewed primary LLM-assisted input did not change the current primary pooled-correlation subset.

Not permitted yet: full-corpus M1-R accuracy, all-construct/all-row SEM substitution stability, autonomous LLM replacement, or vendor ranking.

Team Writing Brief: Literature Review and Discussion

Date: 2026-06-12

Lead owner: Hosung. Lead retains Introduction, Methods, Results, analysis execution, claim boundary, and final integration. Team members should draft only Literature Review and Discussion materials.

Paper A Literature Review Assignment

Target length: 1,500 to 2,000 words.

Deliverables:

- Define the 10 constructs: PE, EE, SI, FC, ATT, SE, TRU, ANX, BI, UB.
- Synthesize TAM, UTAUT, TPB/TRA, and AI-specific trust/anxiety literature.
- Explain why attitude remains theoretically important in higher-education AI adoption.
- Position the paper against Scherer et al. and broader education-technology MASEM work.
- Draft hypotheses or research questions without claiming final path estimates.

Do not write:

- Do not report Stage 1 or Stage 2 estimates.
- Do not imply every construct pair has strong coverage.

- Do not treat converted beta/path values as direct correlations.

Paper A Discussion Assignment

Target length after results insertion: 1,500 to 2,000 words.

Deliverables:

- Interpret final TSSEM/OSMASEM results after lead inserts estimates.
- Discuss theory implications for TAM/UTAUT in AI-specific higher education.
- Discuss practical implications for institutions adopting generative or adaptive AI tools.
- Discuss limitations: missing sample size, incomplete matrices, source-type sensitivity, construct heterogeneity, publication bias.
- Keep implications conditional until final estimates are inserted.

Paper B Literature Review Assignment

Target length: 1,500 to 2,000 words.

Deliverables:

- Review AI/LLM-assisted data extraction in systematic reviews and meta-analyses.
- Emphasize source-anchored reference standards, locked outputs, prompt/model versioning, and human-in-the-loop validation.
- Explain why MASEM extraction is harder than simple metadata extraction.
- Use task-family and downstream-consequence language rather than one overall denominator.
- Position the study in RSM/evidence-synthesis methodology, not only education.

Do not write:

- Do not say gold standard.
- Do not say the LLM replaces human coders.
- Do not rank Codex, Claude, and Gemini as vendors in the main claim.
- Do not treat 8,783 task units as one accuracy denominator.
- Do not call the 90-row M1-R shard a full-corpus result.

Paper B Discussion Assignment

Target length after final gate decision: 1,500 to 2,000 words.

Deliverables:

- Interpret augmentation value, review triage, and source-risk visibility.
- Explain where the workflow failed: high-consequence numeric extraction and abstention.
- Discuss why task-family denominators and source conditions matter for research synthesis.
- Discuss OSF/public repository contribution and closed-PDF/private workbook boundaries.
- State next-method implications for evidence synthesis teams adopting LLM workflows.

Shared Reference Set for Team Authors

- Computers & Education guide: <https://www.sciencedirect.com/journal/computers-and-education/publish/guide-for-authors>
- RSM instructions: <https://www.cambridge.org/core/journals/research-synthesis-methods/information/author-instructions/preparing-your-materials>
- Scherer et al. DOI: <https://doi.org/10.1016/j.compedu.2018.09.009>
- Or (2024) DOI: <https://doi.org/10.18357/otessaj.2024.4.3.66>
- Schmidt et al. arXiv: <https://arxiv.org/abs/2405.14445>
- Schroeder et al. arXiv: <https://arxiv.org/abs/2501.11840>
- RSM GenAI guidance DOI: <https://doi.org/10.1017/rsm.2025.10058>
- Paper B OSF: <https://osf.io/mkrgd/overview>

Submission Readiness Gap Register

Date: 2026-06-12

P0 Gates

Paper	Gate	Current state	Required before submission claim
Paper A	N-weighted TSSEM/OSMASEM	754 of 796 usable primary rows lack numeric N	Reconcile sample sizes from source-supported records or approve an N-eligible subset rule specific to Paper A
Paper A	Final inclusion count	Proposal count and working final count are not harmonized in the manuscript	Lock final PRISMA counts and final MASEM-eligible study count
Paper A	Final Stage 1/Stage 2 estimates	Not generated in repo-local submission package	Run 10-construct TSSEM, sensitivity, and moderator analyses after N gate
Paper A	Closed target exemplar PDF	Scherer C&E PDF not available in share-safe workspace	Use PSU access PDF locally for final exact table/figure comparison if needed
Paper B	Full-corpus M1-R expansion	10 private source packets present; 194 expected from the full-coverage gate	Restore or regenerate all private source packets, run full 2,043-row M1-R, then exception-aware scoring
Paper B	Full-corpus accuracy claim	90-row M1-R shard only	Do not claim full-corpus accuracy until the full run is locked and scored
Paper B	All-row SEM substitution stability	Bounded core-6 subset diagnostic only	Complete final TSSEM/MASEM specification and N/source eligibility before all-row claim
Paper B	Closed comparator PDFs	Annals/RSM exact PDF layouts not all	Use PSU access PDFs for exact final

Paper	Gate	Current state	Required before submission claim
		locally available	table/figure deconstruction if target formatting requires it

P1 Build Tasks

- Generate Paper A PRISMA diagram after final inclusion lock.
- Generate Paper A coverage heatmap and conceptual model figure.
- Generate Paper B workflow, scoring architecture, model-provenance, and triage figures.
- Convert table spine into final journal-ready Word tables.
- Run citation manager cleanup and journal reference style conversion.
- Update OSF data-availability text after any new share-safe outputs are added.